

Growth vs. Sustainability: Can We Predict the Tradeoff?

You have joined a policy analytics team advising an international climate consortium. Global leaders are preparing for the next round of climate agreements, and they need data-driven insights to guide decisions. You are tasked to investigate one of the most debated questions in modern economics and environmental science:

Does economic growth inevitably come at the cost of increased carbon emissions?

You've been given access to historical data on GDP per capita and CO₂ emissions per capita for several countries. Some are rapidly developing. Others are highly industrialized. Each tells a different story. But this isn't just about looking at past trends. Your team needs forward-looking insights. Can the relationship between wealth and emissions help predict the future? Will growing economies continue to emit more carbon, or is there evidence of change?

The deliverable:

As a data analyst, you are tasked with producing a comprehensive data-driven report and forecasting analysis that evaluates both the historical relationship and future trajectory of GDP per capita and CO₂ emissions per capita.

To achieve this, you will:

- Analyze and visualize the relationship between economic growth and emissions across multiple countries
- Determine whether a consistent relationship exists, and how it varies by country
- Apply time-series forecasting models, specifically ARIMA, SARIMA, and SARIMAX, to predict future GDP per capita and CO₂ emissions per capita
- Evaluate the performance and reliability of these models

Your final deliverable should include:

- A clear, well-structured analysis of trends and relationships in the data
- A forecasting component using ARIMA, SARIMA, and SARIMAX models to generate future projections
- Visualizations that effectively communicate both historical patterns and predicted outcomes
- A nuanced conclusion of whether economic growth and emissions are inherently linked, supported by your results and model performance

Your work will be presented as if to policymakers, meaning your results must be not only technically sound, but also clearly communicated and actionable.

Some useful references:

The relationship between growth in GDP and CO₂ has loosened; it needs to be cut completely:

<https://www.iea.org/commentaries/the-relationship-between-growth-in-gdp-and-co2-has-loosened-it-needs-to-be-cut-completely>

ARIMA vs. SARIMA vs. SARIMAX:

<https://medium.com/@sophiamsac/arima-vs-sarima-vs-sarimax-03dd04fc7c66>